# The Costs of Production

EC 111 Principles of Microeconomics

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# Reading

- These slides are not a substitute for reading the textbook.
- Required reading: Mankiw, *Principles of Microeconomics*, 11th edition, Chapter 14, "The Costs of Production".

# What you must know from this chapter

1. Economic cost versus accounting cost.
2. Diminishing marginal product.
3. Short-run total, average and marginal cost.
4. The relationship between marginal cost and marginal product.
5. The relationship between marginal cost and average costs.
6. Long-run average cost.
7. Economies and diseconomies of scale.

Part 1

# Economic Cost versus Accounting Cost

# Explicit, implicit, and economic costs

**Definition: Explicit cost**

A payment the firm makes, such as wages, rent, or materials.

**Definition: Implicit cost**

The value of a resource used by the firm that could have been used elsewhere, even though no money changes hands. The owner's forgone salary and forgone interest on invested savings are examples.

**Definition: Economic cost**

The opportunity cost of all resources used in production: explicit costs *plus* implicit costs.

An accountant ordinarily records payments made by the business. An economist also counts the income the owner sacrifices by using her own labor and capital in the firm.

# Accounting profit and economic profit

$$\text{Accounting profit} = \text{Total revenue} - \text{Explicit costs}$$

$$\text{Economic profit} = \text{Total revenue} - \text{Explicit costs} - \text{Implicit costs}$$

**Finding**

Economic profit equals accounting profit *minus* implicit costs. Economic profit of zero means that the firm covers all costs, including the owner's opportunity costs. It is not the same as earning nothing.

# Your turn: an engineer starts a business

An engineer gives up a $120,000 annual salary and invests $500,000 of savings that could earn 5% annually. The business earns revenue of $1,000,000 and pays $800,000 in explicit costs.

1. What is the accounting profit?
2. What are the implicit costs?
3. What is the economic profit?

# Solution: the engineer's business

1. **Accounting profit** = revenue − explicit costs

$$\$1{,}000{,}000 - \$800{,}000 = \$200{,}000$$

2. **Implicit costs** = forgone salary + forgone interest

$$\$120{,}000 + (0.05 \times \$500{,}000) = \$120{,}000 + \$25{,}000 = \$145{,}000$$

3. **Economic profit** = accounting profit − implicit costs

$$\$200{,}000 - \$145{,}000 = \$55{,}000$$

The business earns \$55,000 more than the owner's specified next-best alternatives. Positive accounting profit alone would not establish that conclusion.

# Check: Part 1

A firm's accounting-profit calculation omits its implicit opportunity costs.

**True or false?**

# Check: Part 1 – answer

A firm's accounting-profit calculation omits its implicit opportunity costs.

**Answer: True**

**Finding**

Accounting profit subtracts explicit costs from total revenue. Implicit costs are opportunity costs without a cash outlay by the firm, so they are omitted from accounting profit.

# Part 1 summary: economic cost versus accounting cost

- **Explicit cost:** a payment the firm makes, such as wages, rent or materials.
- **Implicit cost:** the value of a resource the firm uses that could have been used elsewhere, such as the owner's forgone salary and forgone interest.
- Accounting profit = revenue − explicit costs. Economic profit = revenue − explicit costs − implicit costs.
- Zero economic profit means the firm covers all its costs, including the owner's opportunity costs.

Part 2

# Production: The Law of Diminishing Marginal Productivity

# Production function

**Definition: Production function**

The maximum output a firm can produce from specified quantities of inputs, given its technology. In the **short run**, at least one input is fixed. With a fixed factory (capital $\bar{K}$) and variable labor $L$:

$$Q = f(L, \bar{K}).$$

It shows how output changes as the firm adds workers to the same factory.

**Definition: Marginal and average product**

The **marginal product of labor (MP)** is the additional output obtained from an additional unit of labor. The **average product of labor (AP)** is output per unit of labor:

$$\text{MP} = \frac{\Delta Q}{\Delta L}, \qquad \text{AP} = \frac{Q}{L} \quad (L > 0).$$

# Output from a fixed factory

| Workers $L$ | Output $Q$ | Added output (MP) | AP $= Q/L$ |
|---|---|---|---|
| 0 | 0 | — | — |
| 1 | 18 | 18 $= 18 - 0$ | 18 $= 18 \div 1$ |
| 2 | 48 | 30 $= 48 - 18$ | 24 $= 48 \div 2$ |
| 3 | 84 | 36 $= 84 - 48$ | 28 $= 84 \div 3$ |
| 4 | 120 | 36 $= 120 - 84$ | 30 $= 120 \div 4$ |
| 5 | 150 | 30 $= 150 - 120$ | 30 $= 150 \div 5$ |
| 6 | 168 | 18 $= 168 - 150$ | 28 $= 168 \div 6$ |
| 7 | 168 | 0 $= 168 - 168$ | 24 $= 168 \div 7$ |

The MP column is the increase in output when that worker is hired.

The table shows a first phase of increasing MP, followed by diminishing MP. Eventually, the extra worker adds nothing.

# The law of diminishing marginal productivity

In the table, MP is 18, 30, 36, 36, 30, 18, 0.

**Finding**

**Law of diminishing marginal productivity.** As more labor is added while the factory and other inputs are held fixed, the extra output produced by an additional worker *eventually falls*. MP need not fall from the first worker. Initially, specialization can make MP rise; later, workers have to share the same fixed equipment and space.

| $L$ | $Q$ | MP | AP |
|---|---|---|---|
| 1 | 18 | 18 | 18 |
| 2 | 48 | 30 | 24 |
| 3 | 84 | 36 | 28 |
| 4 | 120 | 36 | 30 |
| 5 | 150 | 30 | 30 |
| 6 | 168 | 18 | 28 |
| 7 | 168 | 0 | 24 |

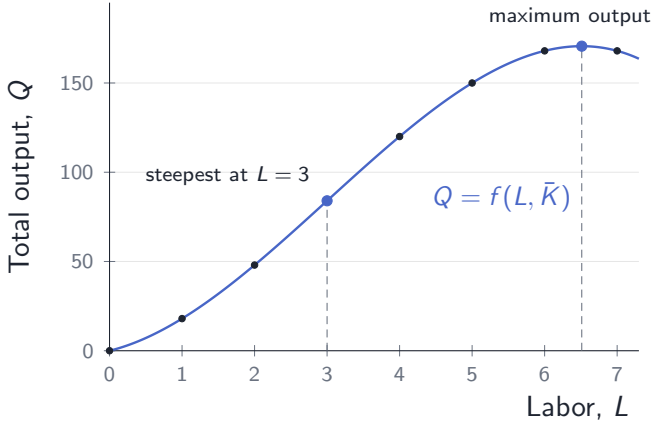
# Total product

Function: $Q = 10L + 9L^2 - L^3$.

The curve is the notes' production function; the dots are the table.

Total product first rises at an increasing rate. It is steepest at $L = 3$.

It then rises at a decreasing rate, and reaches its maximum when the extra output from another small increase in labor has fallen to zero: about 170.6 units at $L \approx 6.5$.



# Marginal product and average product

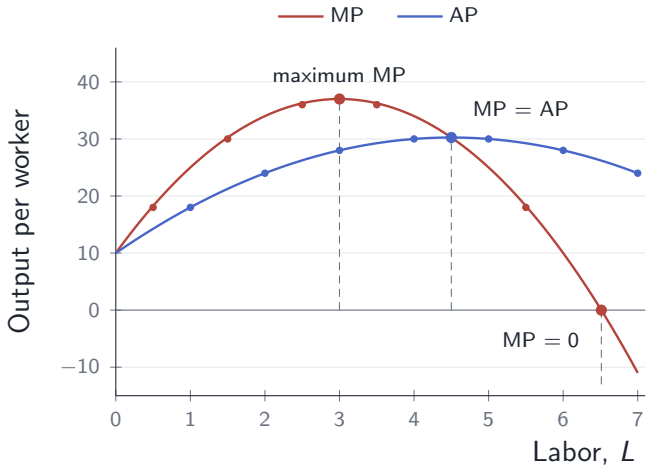
Functions: $\text{AP} = Q/L = 10 + 9L - L^2$; smooth $\text{MP} = 10 + 18L - 3L^2$.

For this smooth example, MP is greatest at $L = 3$ (MP $= 37$).

AP is greatest at $L = 4.5$, where MP $=$ AP $= 30.3$.

MP becomes zero when total output reaches its peak, at $L \approx 6.5$.

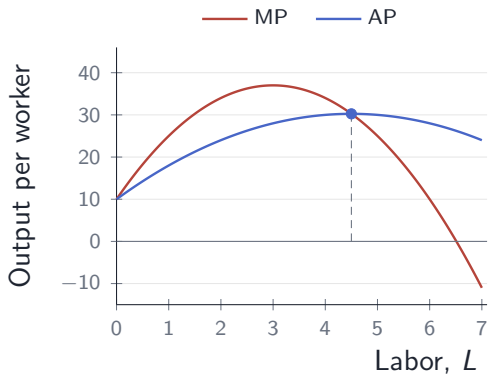
Dots are the table. MP dots sit halfway between worker numbers: the 5th worker's 30 is at 4.5.



# Marginal product and average product

**Finding**

When MP $>$ AP, average product rises. When MP $<$ AP, average product falls. On the usual smooth, single-peaked curves, MP crosses AP *from above* at the maximum of AP. A new worker producing more than the current average raises output per worker; a worker producing less pulls it down.



# Check: Part 2

A print shop produces 240 posters a day with 4 workers and 285 posters a day with 5 workers. The marginal product of the fifth worker is

A 45 posters.

B 57 posters.

C 60 posters.

D 285 posters.

# Check: Part 2 – answer

A print shop produces 240 posters a day with 4 workers and 285 posters a day with 5 workers. The marginal product of the fifth worker is

**A 45 posters.**

B 57 posters.

C 60 posters.

D 285 posters.

**Finding**

The marginal product of a worker is the increase in output from hiring that worker: $285 - 240 = 45$ posters. 57 posters is output per worker with five workers ($285 \div 5$), and 60 is output per worker with four ($240 \div 4$); both are averages, not the change. 285 is total output.

# Part 2 summary: production and diminishing marginal product

- The production function gives the maximum output from given inputs. In the short run, at least one input (the factory) is fixed.
- **MP** = extra output from one more worker = $\Delta Q/\Delta L$. **AP** = output per worker = $Q/L$.
- **Diminishing marginal product:** with the factory fixed, the extra output from another worker eventually falls.
- When MP $>$ AP, AP rises. When MP $<$ AP, AP falls. MP crosses AP at the maximum of AP.

Part 3

# Short-Run Costs: Total, Average, and Marginal

# Total, fixed, and variable cost

**Definition: Fixed cost (FC)**

Does not change when output changes in the short run, such as a factory lease. Fixed cost is incurred even when the firm produces nothing, provided the cost cannot be avoided during the period considered.

**Definition: Variable cost (VC)**

Changes with output, such as the cost of labor or materials.

**Definition: Total cost (TC)**

The sum of fixed and variable cost: $\text{TC} = \text{FC} + \text{VC}$.

The example: a firm with fixed cost \$100. Its variable cost depends on how many units $Q$ it makes.

# The firm's total costs

| $Q$ | FC | VC | TC = FC + VC |
|---|---|---|---|
| 0 | \$100 | \$0 | \$100 = 100 + 0 |
| 1 | \$100 | \$22 | \$122 = 100 + 22 |
| 2 | \$100 | \$32 | \$132 = 100 + 32 |
| 3 | \$100 | \$36 | \$136 = 100 + 36 |
| 4 | \$100 | \$40 | \$140 = 100 + 40 |
| 5 | \$100 | \$50 | \$150 = 100 + 50 |
| 6 | \$100 | \$72 | \$172 = 100 + 72 |
| 7 | \$100 | \$112 | \$212 = 100 + 112 |
| 8 | \$100 | \$176 | \$276 = 100 + 176 |

A new example, separate from the factory in Part 2. This firm's variable cost is $\text{VC} = Q^3 - 9Q^2 + 30Q$. For example, $Q = 2$: $8 - 36 + 60 = \$32$.

Fixed cost is \$100 in every row, including $Q = 0$.

Variable cost is \$0 at $Q = 0$ and rises with output.

Total cost adds the two.

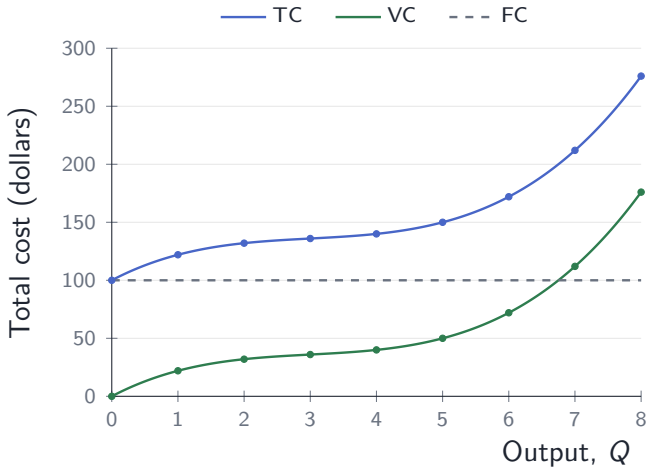
# Total cost, variable cost and fixed cost

Functions: $\text{TC} = Q^3 - 9Q^2 + 30Q + 100$; $\text{VC} = Q^3 - 9Q^2 + 30Q$; $\text{FC} = 100$.

FC is horizontal.

TC is always FC above VC: the distance between TC and VC stays at \$100 at every output.

The total-cost curve becomes increasingly steep at higher output because marginal cost is rising.



# Average costs and marginal cost

**Definition: Average costs**

Average cost means *cost per unit of output*. For positive output,

$$\text{AFC} = \frac{\text{FC}}{Q}, \qquad \text{AVC} = \frac{\text{VC}}{Q}, \qquad \text{ATC} = \frac{\text{TC}}{Q} = \text{AFC} + \text{AVC}.$$

**Definition: Marginal cost (MC)**

The additional cost of producing additional output:

$$\text{MC} = \frac{\Delta \text{TC}}{\Delta Q} = \frac{\Delta \text{VC}}{\Delta Q}.$$

The second equality follows because fixed cost does not change when output changes.

In the tables, output rises one unit at a time, so MC is this row's TC minus the row above's TC.

# The full cost table

| $Q$ | TC | AFC<br>$100 \div Q$ | AVC<br>$VC \div Q$ | ATC<br>$TC \div Q$ | MC<br>change in TC |
|---|---|---|---|---|---|
| 0 | \$100 | — | — | — | — |
| 1 | \$122 | \$100.0 | \$22 | \$122.0 | \$22 $= 122 - 100$ |
| 2 | \$132 | \$50.0 | \$16 | \$66.0 | \$10 $= 132 - 122$ |
| 3 | \$136 | \$33.3 | \$12 | \$45.3 | \$4 $= 136 - 132$ |
| 4 | \$140 | \$25.0 | \$10 | \$35.0 | \$4 $= 140 - 136$ |
| 5 | \$150 | \$20.0 | \$10 | \$30.0 | \$10 $= 150 - 140$ |
| 6 | \$172 | \$16.7 | \$12 | \$28.7 | \$22 $= 172 - 150$ |
| 7 | \$212 | \$14.3 | \$16 | \$30.3 | \$40 $= 212 - 172$ |
| 8 | \$276 | \$12.5 | \$22 | \$34.5 | \$64 $= 276 - 212$ |

Check $Q = 5$: AFC $= 100 \div 5 =$ \$20.0; AVC $= 50 \div 5 =$ \$10; ATC $= 150 \div 5 =$ \$30.0 $=$ AFC $+$ AVC; MC $= 150 - 140 =$ \$10.

# Your turn: calculating costs

Fill in the blanks. One entry in each row is given (bold).

| $Q$ | VC | TC | AFC | AVC | ATC | MC |
|---|---|---|---|---|---|---|
| 0 | \$0 | \$100 | — | — | — | — |
| 1 | | **\$122** | | | | |
| 2 | | | | **\$16** | | |
| 3 | | | | | | **\$4** |
| 4 | **\$40** | | | | | |
| 5 | | | | | **\$30.0** | |
| 6 | | | | | | **\$22** |

Hint: start with fixed cost, which is TC at $Q = 0$.

# Solution: calculating costs

First, FC = TC at $Q = 0$ = \$100. Then use TC = FC + VC, AFC = FC $\div Q$, AVC = VC $\div Q$, ATC = TC $\div Q$, and MC = change in TC.

| $Q$ | VC | TC | AFC | AVC | ATC | MC |
|---|---|---|---|---|---|---|
| 0 | \$0 | \$100 | — | — | — | — |
| 1 | \$22 | **\$122** | \$100.0 | \$22 | \$122.0 | \$22 |
| 2 | \$32 | \$132 | \$50.0 | **\$16** | \$66.0 | \$10 |
| 3 | \$36 | \$136 | \$33.3 | \$12 | \$45.3 | **\$4** |
| 4 | **\$40** | \$140 | \$25.0 | \$10 | \$35.0 | \$4 |
| 5 | \$50 | \$150 | \$20.0 | \$10 | **\$30.0** | \$10 |
| 6 | \$72 | \$172 | \$16.7 | \$12 | \$28.7 | **\$22** |

Example, row 3: MC = \$4 means TC = 132 + 4 = \$136, so VC = 136 − 100 = \$36.

# Fixed cost versus average fixed cost

Total fixed cost stays constant as output rises. Spread \$100 over more units:

$$\frac{100}{2} = \$50.0, \quad \frac{100}{4} = \$25.0, \quad \frac{100}{8} = \$12.5.$$

**Finding**

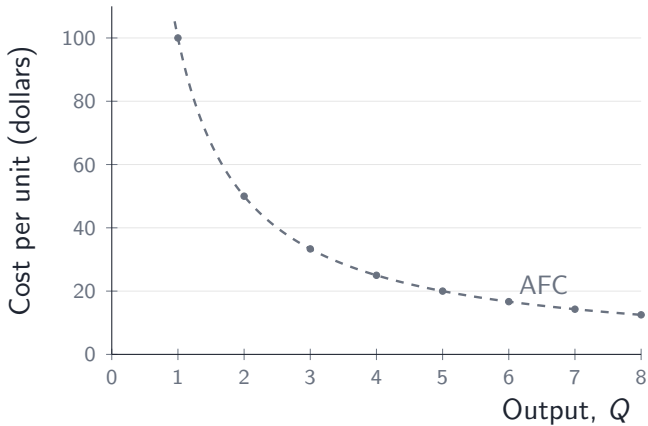
Total fixed cost stays constant as output rises. But average fixed cost always falls for $Q > 0$, because the same fixed sum is spread over more units. Do not confuse constant FC with falling AFC.

# Average fixed cost

Function: $\text{AFC} = \text{FC}/Q = 100/Q$.

AFC declines throughout: the same \$100 is spread over more units.

Dots are the table.



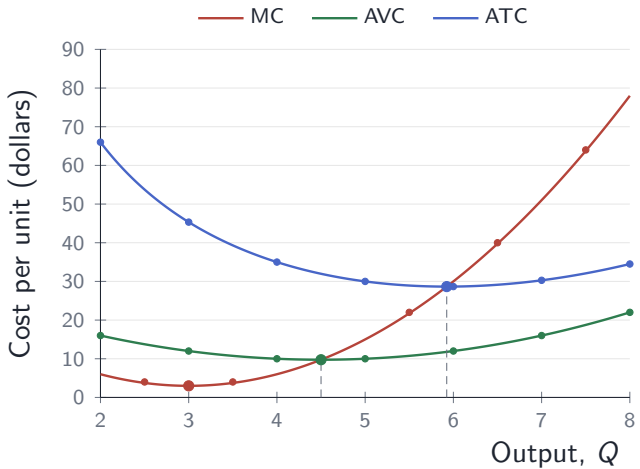
# Short-run unit costs

Functions: $\text{MC} = 3Q^2 - 18Q + 30$; $\text{AVC} = Q^2 - 9Q + 30$; $\text{ATC} = \text{AVC} + 100/Q$.

MC is lowest first, at $Q = 3$.

MC meets AVC at its minimum ($Q = 4.5$) and ATC at its minimum ($Q \approx 5.9$). These intersections are marked by dots.

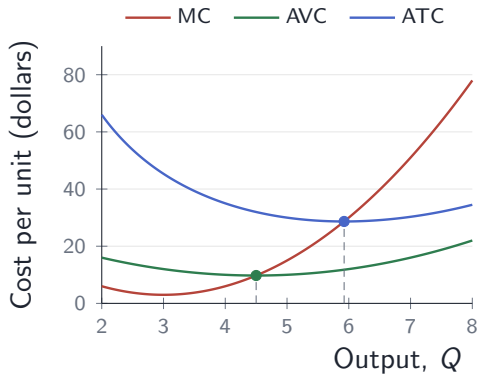
Small dots are the table. MC dots sit halfway between the two outputs compared.



# Short-run unit costs

**Finding**

AFC falls continuously. In the usual short-run model, MC falls at first and then rises as diminishing marginal productivity sets in. AVC and ATC typically become U-shaped. The minimum of MC occurs before the minima of AVC and ATC.



# Check: Part 3

A bakery pays each month for flour, the wages of hourly counter staff, electricity to run its ovens, and the lease on its shop. The counter staff work more hours when output rises. Which of these is a fixed cost?

A The flour

B The wages of hourly counter staff

C The electricity to run its ovens

D The lease on its shop

# Check: Part 3 – answer

A bakery pays each month for flour, the wages of hourly counter staff, electricity to run its ovens, and the lease on its shop. The counter staff work more hours when output rises. Which of these is a fixed cost?

A The flour

B The wages of hourly counter staff

C The electricity to run its ovens

**D The lease on its shop**

**Finding**

A fixed cost does not change with the quantity of output. The lease is the same whether the bakery bakes 100 loaves a month or 10,000. Flour, hourly wages and oven electricity all rise as the bakery produces more, so they are variable costs.

# Part 3 summary: short-run total, average and marginal cost

- **FC** does not change with output. **VC** changes with output. TC = FC + VC.
- AFC = FC/$Q$, AVC = VC/$Q$, ATC = TC/$Q$, and ATC = AFC + AVC.
- **MC** = extra cost of one more unit of output = $\Delta$TC/$\Delta Q$.
- AFC always falls. MC falls at first, then rises. AVC and ATC are U-shaped.

Part 4

# The Link between Marginal Cost and Marginal Product

# Back to the factory

- This part uses the factory from Part 2. It is not the cost table of Part 3.
- Production function: $Q = 10L + 9L^2 - L^3$.
- Labor is the only variable input. Every worker is paid \$36.

# Output from each number of workers

| Workers $L$ | Output $Q = 10L + 9L^2 - L^3$ |
|---|---|
| 1 | $10 + 9 - 1 = 18$ |
| 2 | $20 + 36 - 8 = 48$ |
| 3 | $30 + 81 - 27 = 84$ |
| 4 | $40 + 144 - 64 = 120$ |
| 5 | $50 + 225 - 125 = 150$ |
| 6 | $60 + 324 - 216 = 168$ |

These are the same outputs as the Part 2 table.

# Cost of each number of workers

Each worker costs \$36, so cost $C = 36 \times L$.

| Workers $L$ | Output $Q$ | Cost $C = 36 \times L$ |
|---:|---:|---:|
| 1 | 18 | \$36 |
| 2 | 48 | \$72 |
| 3 | 84 | \$108 |
| 4 | 120 | \$144 |
| 5 | 150 | \$180 |
| 6 | 168 | \$216 |

# Read the table backwards

- To make 84 units, the firm needs 3 workers.
- Cost is $36 \times 3 = \$108$.
- Going from output back to workers is the **inverse** of the production function.

**Finding**

Production determines how much labor is needed for a given output. If the firm hires $L$ workers, its variable cost is $wL$, with wage $w = \$36$.

| Workers $L$ | Output $Q$ | Cost $C$ |
|---|---|---|
| 1 | 18 | \$36 |
| 2 | 48 | \$72 |
| 3 | 84 | \$108 |
| 4 | 120 | \$144 |
| 5 | 150 | \$180 |
| 6 | 168 | \$216 |

# Marginal cost from the table

$\text{MC} = \Delta C/\Delta Q$: the extra cost of one more unit of **output**.

| Move | $\Delta C$ | $\Delta Q$ | $\text{MC} = \Delta C/\Delta Q$ |
|---|---|---|---|
| $0 \to 1$ | \$36 | 18 | \$2.0 |
| $1 \to 2$ | \$36 | 30 | \$1.2 |
| $2 \to 3$ | \$36 | 36 | \$1.0 |
| $3 \to 4$ | \$36 | 36 | \$1.0 |
| $4 \to 5$ | \$36 | 30 | \$1.2 |
| $5 \to 6$ | \$36 | 18 | \$2.0 |

# What the table shows

- $\Delta C$ is \$36 in every row: one more worker costs one wage.
- $\Delta Q$ is the extra output from that worker, which is MP.
- So MC $=$ \$36 $\div$ MP.

| Move | $\Delta C$ | $\Delta Q$ | MC |
|---|---|---|---|
| $0 \to 1$ | \$36 | 18 | \$2.0 |
| $1 \to 2$ | \$36 | 30 | \$1.2 |
| $2 \to 3$ | \$36 | 36 | \$1.0 |
| $3 \to 4$ | \$36 | 36 | \$1.0 |
| $4 \to 5$ | \$36 | 30 | \$1.2 |
| $5 \to 6$ | \$36 | 18 | \$2.0 |

# MC is not the wage

| | Extra cost of... | Formula | Here |
|---|---|---|---|
| **MC** | one more unit of **output** | $\Delta C/\Delta Q$ | \$2.0, 1.2, 1.0, ... |
| **Wage** | one more **worker** | $\Delta C/\Delta L$ | \$36 |

# Linking MC and the wage

$$\underbrace{\frac{\Delta C}{\Delta Q}}_{\text{MC}} = \underbrace{\frac{\Delta C}{\Delta L}}_{\text{wage}} \times \underbrace{\frac{\Delta L}{\Delta Q}}_{1/\text{MP}}$$

One worker adds MP units, so one unit needs $1/\text{MP}$ of a worker (because $\Delta L$ is 1 and $\Delta Q$ is MP).

# Check with the 3rd worker

- $\Delta L = 3 - 2 = 1$.
- $\Delta Q = 84 - 48 = 36 = \text{MP}$.
- $\text{MC} = \$36 \times 1/36 = \$1.0$. This is the same as the table.

**Finding**

MC = wage $\div$ MP.

| Workers $L$ | Output $Q$ | Cost $C$ |
|---|---|---|
| 1 | 18 | \$36 |
| 2 | 48 | \$72 |
| 3 | 84 | \$108 |
| 4 | 120 | \$144 |
| 5 | 150 | \$180 |
| 6 | 168 | \$216 |

# Average variable cost from average product

$$\text{AVC} = \frac{wL}{Q} = \frac{w}{Q/L} = \frac{w}{\text{AP}}$$

Check at 3 workers: $\$108 \div 84 = \$1.3$, and AP $= 84 \div 3 = 28$, so $\$36 \div 28 = \$1.3$.

**Finding**

AVC = wage ÷ AP.

| Workers $L$ | Output $Q$ | Cost $C$ |
|---|---|---|
| 1 | 18 | \$36 |
| 2 | 48 | \$72 |
| 3 | 84 | \$108 |
| 4 | 120 | \$144 |
| 5 | 150 | \$180 |
| 6 | 168 | \$216 |

# The full table

| $L$ | $Q$ | MP | MC $= 36 \div$ MP | AP | AVC $= 36 \div$ AP |
|---|---|---|---|---|---|
| 1 | 18 | 18 | \$2.0 | 18 | \$2.0 |
| 2 | 48 | 30 | \$1.2 | 24 | \$1.5 |
| 3 | 84 | 36 | \$1.0 | 28 | \$1.3 |
| 4 | 120 | 36 | \$1.0 | 30 | \$1.2 |
| 5 | 150 | 30 | \$1.2 | 30 | \$1.2 |
| 6 | 168 | 18 | \$2.0 | 28 | \$1.3 |

# What the full table shows

- As MP rises from 18 to 36, MC falls from \$2.0 to \$1.0.
- As MP falls, MC rises.
- AVC is lowest (\$1.2) where AP is highest (30).

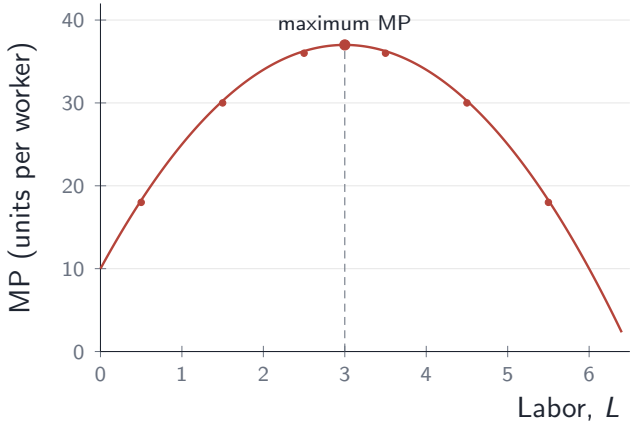
| $L$ | $Q$ | MP | MC | AP | AVC |
|---|---|---|---|---|---|
| 1 | 18 | 18 | \$2.0 | 18 | \$2.0 |
| 2 | 48 | 30 | \$1.2 | 24 | \$1.5 |
| 3 | 84 | 36 | \$1.0 | 28 | \$1.3 |
| 4 | 120 | 36 | \$1.0 | 30 | \$1.2 |
| 5 | 150 | 30 | \$1.2 | 30 | \$1.2 |
| 6 | 168 | 18 | \$2.0 | 28 | \$1.3 |

# Marginal product against labor

Function: $\text{MP} = 10 + 18L - 3L^2$.

MP rises, peaks at $L = 3$ with $\text{MP} = 37$, then falls.

Dots are the table, halfway between worker numbers.



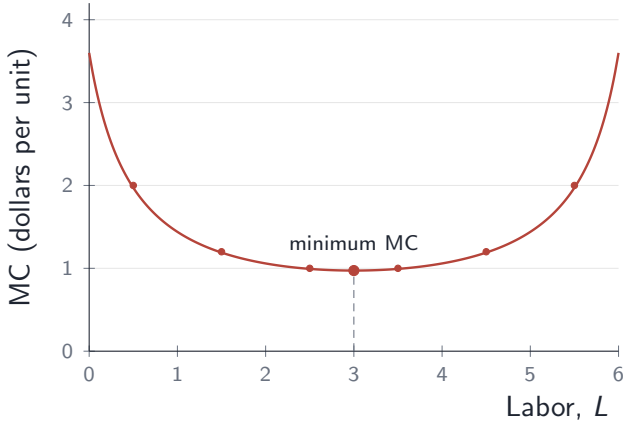
# Marginal cost against labor

Function: $\text{MC} = 36/(10 + 18L - 3L^2)$.

The vertical axis is MC, dollars per extra unit of **output**. The horizontal axis is labor, so this graph lines up with the MP graph.

MC is lowest at $L = 3$: $\$36 \div 37 = \$1.0$.

Dots are the table, halfway between worker numbers.



# MC moves opposite to MP

- MP $\uparrow \Rightarrow$ MC $\downarrow$. MP $\downarrow \Rightarrow$ MC $\uparrow$.
- Maximum MP gives minimum MC.
- Maximum AP gives minimum AVC.

# Your turn: marginal cost from marginal product

The wage is \$36.

| Workers $L$ | Output $Q$ | Cost $C$ |
|---|---|---|
| 2 | 48 | \$72 |
| 3 | 84 | \$108 |
| 5 | 150 | \$180 |
| 6 | 168 | \$216 |

1. From 2 to 3 workers: what are $\Delta C$, $\Delta Q$ and MC?
2. From 5 to 6 workers: what are $\Delta C$, $\Delta Q$ and MC?
3. Which law explains the change?

# Solution: marginal cost from marginal product

| Workers $L$ | Output $Q$ | Cost $C$ |
|---|---|---|
| 2 | 48 | \$72 |
| 3 | 84 | \$108 |
| 5 | 150 | \$180 |
| 6 | 168 | \$216 |

1. $\Delta C = \$108 - \$72 = \$36$; $\Delta Q = 84 - 48 = 36$; MC $= \$36 \div 36 = \$1.0$.
2. $\Delta C = \$216 - \$180 = \$36$; $\Delta Q = 168 - 150 = 18$; MC $= \$36 \div 18 = \$2.0$.
3. Diminishing marginal productivity. Each worker still costs \$36, but the 6th worker adds only 18 units, so each unit costs more.

# Check: Part 4

A call center pays each agent $20 an hour. Each additional agent handles fewer calls than the one hired before. The marginal cost of an additional call rises.

**True or false?**

# Check: Part 4 – answer

A call center pays each agent $20 an hour. Each additional agent handles fewer calls than the one hired before. The marginal cost of an additional call rises.

**Answer: True**

**Finding**

Each extra agent still costs $20 an hour but handles fewer calls, so each additional call costs more. Diminishing marginal product makes marginal cost rise.

# Part 4 summary: marginal cost and marginal product

- **MC** is the extra cost of one more unit of output: $\text{MC} = \Delta C / \Delta Q$.
- **MP** is the extra output from one more worker: $\text{MP} = \Delta Q / \Delta L$.
- With a constant wage, MC = wage $\div$ MP. One more worker costs the wage and produces MP more units.
- MC moves opposite to MP. When MP rises, MC falls. When MP falls, MC rises. The lowest MC comes where MP is highest.
- Diminishing marginal product makes MC rise in the short run.

Part 5

# The Link between Marginal Cost and Average Cost

# Back to the cost table

- This part returns to the cost table of Part 3: $\text{TC} = Q^3 - 9Q^2 + 30Q + 100$. It is not the factory of Part 4.
- Question: when does one more unit raise the average cost, and when does it lower it?

# A marginal grade and a GPA

- After 4 courses, a student's GPA is 3.0. Total grade points: $4 \times 3.0 = 12.0$.
- The 5th course earns 4.0, above 3.0: $(12.0 + 4.0) \div 5 = 3.2$. The GPA **rises**.

# A marginal grade and a GPA (continued)

- The 5th course earns 2.0, below 3.0: $(12.0 + 2.0) \div 5 = 2.8$. The GPA **falls**.
- The 5th course earns 3.0, equal to 3.0: $(12.0 + 3.0) \div 5 = 3.0$. The GPA **stays the same**.

**Finding**

A marginal grade below the average pulls the average down; above it, it pushes the average up. Cost works the same way: MC is the marginal grade and average cost is the GPA.

# The cost table again

| $Q$ | $\text{TC} = Q^3 - 9Q^2 + 30Q + 100$ | $\text{ATC} = \text{TC} \div Q$ | $\text{MC} = \Delta\text{TC}/\Delta Q$ |
|---|---|---|---|
| 4 | $64 - 144 + 120 + 100 = \$140$ | \$35.0 | $(140 - 136)/1 = \$4$ |
| 5 | $125 - 225 + 150 + 100 = \$150$ | \$30.0 | $(150 - 140)/1 = \$10$ |
| 6 | $216 - 324 + 180 + 100 = \$172$ | \$28.7 | $(172 - 150)/1 = \$22$ |
| 7 | $343 - 441 + 210 + 100 = \$212$ | \$30.3 | $(212 - 172)/1 = \$40$ |
| 8 | $512 - 576 + 240 + 100 = \$276$ | \$34.5 | $(276 - 212)/1 = \$64$ |

Output rises one unit at a time, so $\Delta Q = 1$ and $\text{MC} = \Delta\text{TC}/\Delta Q = \Delta\text{TC}$.

# Why ATC falls from 5 to 6 units

- At 5 units, ATC is $\$150 \div 5 = \$30.0$.
- The 6th unit: MC $= \Delta\text{TC}/\Delta Q = (\$172 - \$150)/1 = \$22$. This is **below** the average of \$30.0.
- So ATC falls: $\$172 \div 6 = \$28.7$.

| $Q$ | TC | ATC | MC |
|---|---|---|---|
| 4 | \$140 | \$35.0 | \$4 |
| 5 | \$150 | \$30.0 | \$10 |
| 6 | \$172 | \$28.7 | \$22 |
| 7 | \$212 | \$30.3 | \$40 |
| 8 | \$276 | \$34.5 | \$64 |

# Why ATC rises from 6 to 7 units

- At 6 units, ATC is \$28.7.
- The 7th unit: MC $= (\$212 - \$172)/1 = \$40$. This is **above** the average of \$28.7.
- So ATC rises: $\$212 \div 7 = \$30.3$.

| $Q$ | TC | ATC | MC |
|---|---|---|---|
| 4 | \$140 | \$35.0 | \$4 |
| 5 | \$150 | \$30.0 | \$10 |
| 6 | \$172 | \$28.7 | \$22 |
| 7 | \$212 | \$30.3 | \$40 |
| 8 | \$276 | \$34.5 | \$64 |

# The same rule for AVC

| $Q$ | VC | AVC $=$ VC $\div Q$ | MC |
|---|---|---|---|
| 3 | \$36 | \$12 | \$4 |
| 4 | \$40 | \$10 | \$4 |
| 5 | \$50 | \$10 | \$10 |
| 6 | \$72 | \$12 | \$22 |

- From 3 to 4: MC \$4 is below AVC \$12, so AVC falls to \$10.
- From 4 to 5: MC \$10 equals AVC \$10, so AVC stays at \$10.
- From 5 to 6: MC \$22 is above AVC \$10, so AVC rises to \$12.

# The marginal–average rule

- $\text{MC} < \text{ATC} \Rightarrow$ ATC falls. $\quad \text{MC} > \text{ATC} \Rightarrow$ ATC rises.
- $\text{MC} < \text{AVC} \Rightarrow$ AVC falls. $\quad \text{MC} > \text{AVC} \Rightarrow$ AVC rises.

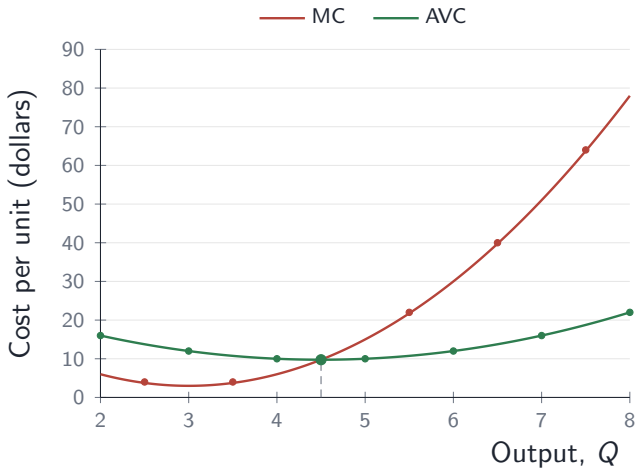
# MC crosses AVC at its minimum

Functions: $\text{MC} = 3Q^2 - 18Q + 30$; $\text{AVC} = Q^2 - 9Q + 30$.

MC crosses AVC from below at $Q = 4.5$, where AVC is lowest: \$9.8.

To the left, MC is below AVC and pulls it down. To the right, MC is above AVC and pushes it up.

Dots are the table. MC dots sit halfway between the two outputs compared.

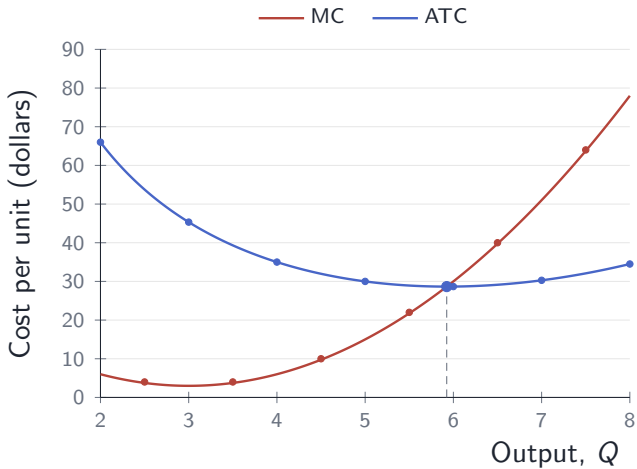


# MC crosses ATC at its minimum

Functions: $\text{MC} = 3Q^2 - 18Q + 30$; $\text{ATC} = Q^2 - 9Q + 30 + 100/Q$.

MC crosses ATC from below at $Q \approx 5.9$, where ATC is lowest: \$28.7.

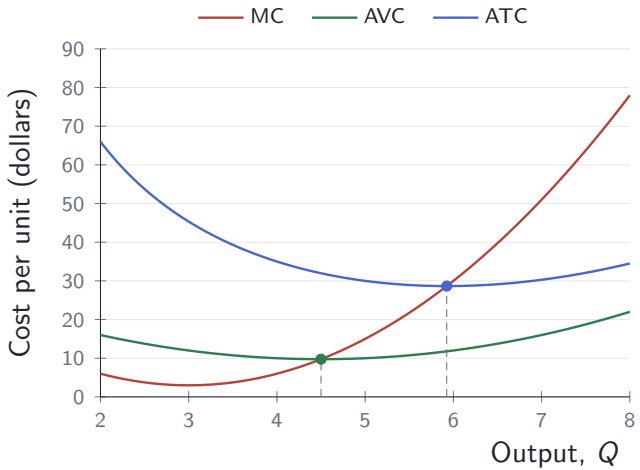
Dots are the table. MC dots sit halfway between the two outputs compared.



# All three together

Functions: $\text{MC} = 3Q^2 - 18Q + 30$; $\text{AVC} = Q^2 - 9Q + 30$; $\text{ATC} = \text{AVC} + 100/Q$.

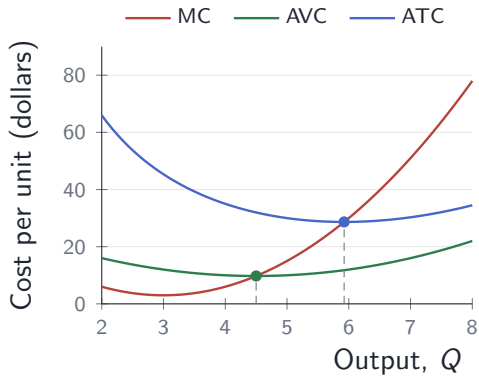
Two crossings and two separate minima: AVC is lowest at $Q = 4.5$ ($9.8); ATC is lowest at $Q \approx 5.9$ ($28.7).



# All three together: finding

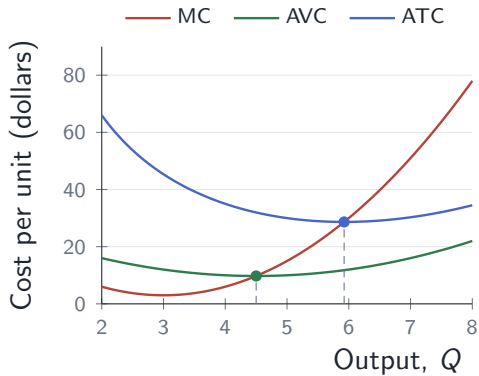
**Finding**

On the usual U-shaped curves, MC crosses AVC and ATC from below at their minima. An equality by itself does not guarantee a minimum; it is the change from MC below the average to MC above it that gives the U-shaped minimum.



# Why the ATC minimum comes after the AVC minimum

- ATC = AVC + AFC.
- At $Q = 4.5$, AVC has reached its minimum, but AFC is still falling.
- So ATC is still falling at 4.5. It reaches its own minimum at a higher output: \$28.7 at $Q \approx 5.9$.



# Your turn: will average total cost rise or fall?

| $Q$ | TC | ATC |
|---|---|---|
| 7 | \$212 | \$30.3 |
| 8 | \$276 | ? |

1. What is the marginal cost of the 8th unit?
2. Without dividing, will ATC rise or fall?
3. Check: what is ATC at 8 units?

# Solution: will average total cost rise or fall?

| $Q$ | TC | ATC |
|---|---|---|
| 7 | \$212 | \$30.3 |
| 8 | \$276 | ? |

1. $\text{MC} = \Delta\text{TC}/\Delta Q = (\$276 - \$212)/(8 - 7) = \$64/1 = \$64$.
2. \$64 is above the average of \$30.3, so ATC rises.
3. $\text{ATC} = \$276 \div 8 = \$34.5$. It rose from \$30.3 to \$34.5.

# Check: Part 5

At an oil refinery, the marginal cost of one more barrel is below the average total cost per barrel. Average total cost falls as the refinery raises output.

**True or false?**

# Check: Part 5 – answer

At an oil refinery, the marginal cost of one more barrel is below the average total cost per barrel. Average total cost falls as the refinery raises output.

**Answer: True**

> **Finding**
> When the extra barrel costs less than the average barrel, producing it pulls the average down. So whenever marginal cost is below average total cost, average total cost is falling.

# Part 5 summary: marginal cost and average cost

- When MC is below an average, it pulls the average down. When MC is above an average, it pushes the average up.
- So MC crosses AVC and ATC from below, at their minima.
- The AVC minimum comes first ($Q = 4.5$, \$9.8). The ATC minimum comes later ($Q \approx 5.9$, \$28.7), because AFC is still falling.

Part 6

# The Long Run: Long-Run Average Cost

# Short run and long run

**Definition: Short run**

At least one input is fixed, so the firm must work with its existing factory or plant.

**Definition: Long run**

All inputs can be changed, including the size of the plant.

# Long-run average cost (LRAC)

**Definition: Long-run average cost (LRAC)**

The lowest possible average total cost of each output when the firm may choose all its inputs, including the plant.

A small plant may be cheap at low output and expensive at high output. A large plant may be the opposite.

# The example

- Long-run average cost: $\text{LRAC} = 30 - 2Q + 0.1Q^2$.
- A plant built for output $s$ has short-run average total cost $\text{ATC}_s = \text{LRAC} + 0.4(Q - s)^2$.
- We draw three plants: small ($s = 5$), medium ($s = 10$) and large ($s = 15$).

# First, LRAC at three outputs

| $Q$ | $\text{LRAC} = 30 - 2Q + 0.1Q^2$ |
|---|---|
| 5 | $30 - 10 + 2.5 = \$22.5$ |
| 10 | $30 - 20 + 10.0 = \$20.0$ |
| 15 | $30 - 30 + 22.5 = \$22.5$ |

# The small plant ($s = 5$)

$\text{ATC}_5 = \text{LRAC} + 0.4(Q - 5)^2$

| $Q$ | LRAC | $+0.4(Q-5)^2$ | $\text{ATC}_5$ |
|---|---|---|---|
| 5 | \$22.5 | $0.4 \times 0 = 0.0$ | \$22.5 |
| 10 | \$20.0 | $0.4 \times 25 = 10.0$ | \$30.0 |
| 15 | \$22.5 | $0.4 \times 100 = 40.0$ | \$62.5 |

# The medium plant ($s = 10$)

$\text{ATC}_{10} = \text{LRAC} + 0.4(Q - 10)^2$

| $Q$ | LRAC | $+0.4(Q-10)^2$ | $\text{ATC}_{10}$ |
|---|---|---|---|
| 5 | \$22.5 | $0.4 \times 25 = 10.0$ | \$32.5 |
| 10 | \$20.0 | $0.4 \times 0 = 0.0$ | \$20.0 |
| 15 | \$22.5 | $0.4 \times 25 = 10.0$ | \$32.5 |

# The large plant ($s = 15$)

$$\text{ATC}_{15} = \text{LRAC} + 0.4(Q - 15)^2$$

| $Q$ | LRAC | $+0.4(Q-15)^2$ | $\text{ATC}_{15}$ |
|---|---|---|---|
| 5 | \$22.5 | $0.4 \times 100 = 40.0$ | \$62.5 |
| 10 | \$20.0 | $0.4 \times 25 = 10.0$ | \$30.0 |
| 15 | \$22.5 | $0.4 \times 0 = 0.0$ | \$22.5 |

# Which plant is cheapest?

| Plant | $Q = 5$ | $Q = 10$ | $Q = 15$ |
|---|---|---|---|
| Small | **$22.5** | $30.0 | $62.5 |
| Medium | $32.5 | **$20.0** | $32.5 |
| Large | $62.5 | $30.0 | **$22.5** |

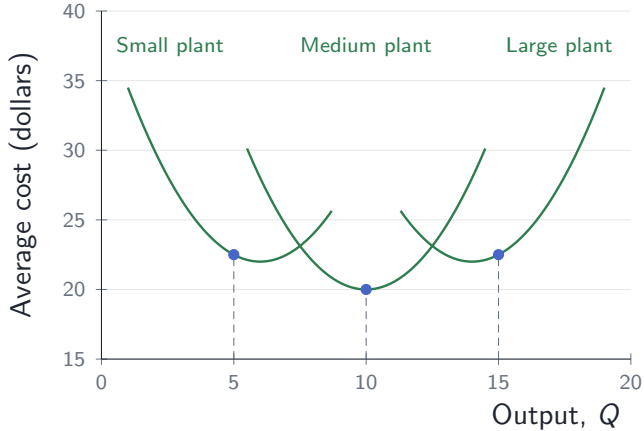
- The lowest cost in each column (bold) shows the best plant for that output.
- Each bold number equals LRAC at that output.

# Three plant sizes

Functions: $\text{ATC}_s = \text{LRAC} + 0.4(Q - s)^2$, $s = 5, 10, 15$; $\text{LRAC} = 30 - 2Q + 0.1Q^2$.

Each plant has its own short-run ATC curve, lowest at the output it was built for.

Dots: (5, \$22.5), (10, \$20.0), (15, \$22.5).



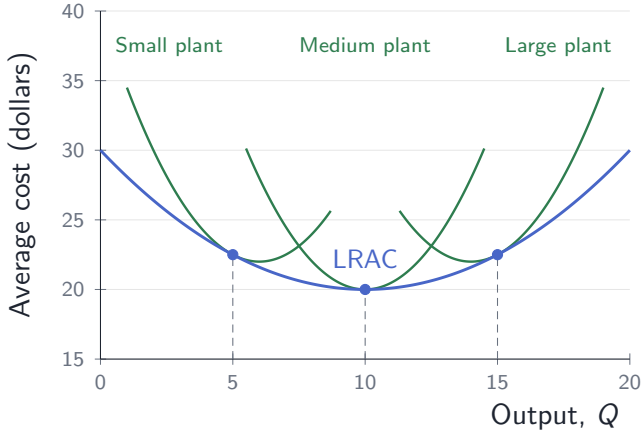
# LRAC is the lower envelope

Functions: $\text{ATC}_s = \text{LRAC} + 0.4(Q - s)^2$, $s = 5, 10, 15$; $\text{LRAC} = 30 - 2Q + 0.1Q^2$.

At every output, LRAC takes the **lowest** cost available among the plants.

Each plant's curve touches LRAC at the output it was built for: $Q = 5$, 10 and 15.

Plants of in-between sizes are also possible; only three are drawn. LRAC is the lower envelope of all of them.



# Your turn: choosing a plant

| Plant | $Q = 5$ | $Q = 10$ | $Q = 15$ |
|---|---|---|---|
| Small | \$22.5 | \$30.0 | \$62.5 |
| Medium | \$32.5 | \$20.0 | \$32.5 |
| Large | \$62.5 | \$30.0 | \$22.5 |

A firm expects to produce 15 units.

1. Which plant should it build in the long run? What is its average cost?
2. It has a medium plant and cannot change it in the short run. What is its average cost at 15 units?
3. How much more per unit does the medium plant cost at 15 units?

# Solution: choosing a plant

| Plant | $Q = 5$ | $Q = 10$ | $Q = 15$ |
|---|---|---|---|
| Small | \$22.5 | \$30.0 | \$62.5 |
| Medium | \$32.5 | \$20.0 | \$32.5 |
| Large | \$62.5 | \$30.0 | \$22.5 |

1. The large plant: \$22.5 per unit.
2. The medium plant: $\$22.5 + 0.4 \times (15 - 10)^2 = \$22.5 + \$10.0 = \$32.5$ per unit.
3. $\$32.5 - \$22.5 = \$10.0$ more per unit. In the long run the firm switches to the large plant.

# Check: Part 6

In the long run, a firm can change all of its inputs, including the size of its factory.

**True or false?**

# Check: Part 6 – answer

In the long run, a firm can change all of its inputs, including the size of its factory.

**Answer: True**

> **Finding**
> In the short run some inputs, such as the factory, are fixed. In the long run the firm has time to change every input, so it can build a larger factory or close one.

# Part 6 summary: long-run average cost

- In the short run, at least one input is fixed. In the long run, the firm can change every input, including the plant.
- Each plant has its own short-run ATC curve.
- LRAC is the lowest average cost at each output when the firm can choose its plant: the lower envelope of the short-run ATC curves.

Part 7

# Economies and Diseconomies of Scale

# Economies and diseconomies of scale

**Definition: Economies of scale**

Occur over a range of outputs when long-run average cost *falls* as output rises.

**Definition: Diseconomies of scale**

Occur over a range when long-run average cost *rises*.

**Definition: Constant average cost**

Over a range with constant average cost, LRAC is flat as output changes.

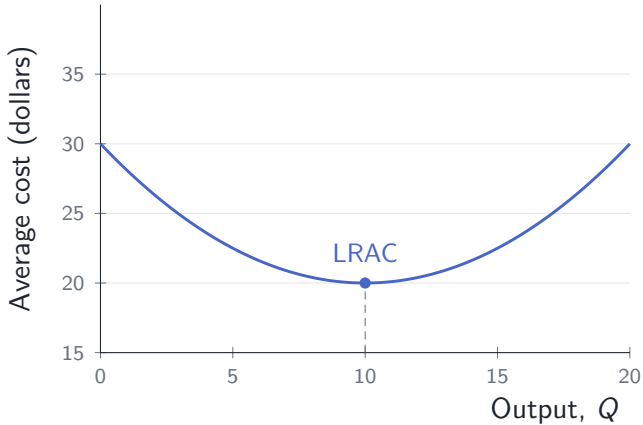
**Definition: Minimum efficient scale (MES)**

The smallest output at which LRAC reaches its lowest level. In the illustrated U-shaped example, MES is $Q = 10$.

# Read the long-run average-cost curve

Function: $\text{LRAC} = 30 - 2Q + 0.1Q^2$.

- **Falling LRAC:** economies of scale, as the firm lowers cost per unit by expanding. Here, from 0 to 10.
- **Flat LRAC over an interval:** constant average costs over that interval.
- **Rising LRAC:** diseconomies of scale, as further expansion raises cost per unit. Here, beyond 10.



# Why LRAC may fall or rise

**Why LRAC may fall.** A larger firm may specialize its labor force more effectively or spread the cost of indivisible equipment over more units.

**Why LRAC may rise.** A very large operation may face increasing costs of communication, supervision, and coordination.

# Do not confuse the short run with the long run

**Finding**

**Diminishing marginal product** is a short-run idea: add more of a variable input while holding another input fixed. It explains why short-run MC eventually rises when the wage is fixed.

**Finding**

**Diseconomies of scale** is a long-run cost idea: LRAC increases as output expands and the firm is free to adjust *all* inputs. Rising short-run MC is not, by itself, evidence of diseconomies of scale.

**Returns to scale** ask what happens to output when *all* inputs are increased in the same proportion.

# Think, pair, share

A market gardener grows produce for a local farmers' market. He hires a summer helper and his output more than doubles. He says: "Next summer I'll hire three helpers, and output should go up three- or fourfold."

1. Why did output more than double with one helper?
2. Will output rise three- or fourfold with three helpers, on the same plot of land?
3. Is this a question about diminishing marginal product or about scale?

# Think, pair, share: discussion

**1.** Specialization: one person can weed while the other waters and picks. Marginal product rose, as in the first workers in the factory table.

**2.** Probably not. The land is fixed. As more helpers share the same plot, each adds less: diminishing marginal product.

**3.** Diminishing marginal product, a short-run idea: one input (land) is fixed. Scale would mean increasing *all* inputs, land included.

# Check: Part 7

A national retail chain grows so large that every decision must pass through several layers of managers. As a result, its long-run average total cost rises as its output grows. The chain is experiencing

A economies of scale.

B constant returns to scale.

C diminishing marginal product.

D diseconomies of scale.

# Check: Part 7 – answer

A national retail chain grows so large that every decision must pass through several layers of managers. As a result, its long-run average total cost rises as its output grows. The chain is experiencing

A economies of scale.

B constant returns to scale.

C diminishing marginal product.

**D diseconomies of scale.**

**Finding**

Diseconomies of scale exist when long-run average total cost rises as output increases. Coordination problems in a very large organization, such as layers of management, are a common cause.

# Part 7 summary: economies and diseconomies of scale

- **Economies of scale:** LRAC falls as output rises.
- **Diseconomies of scale:** LRAC rises as output rises. **Constant average cost:** LRAC is flat.
- **Minimum efficient scale:** the smallest output at which LRAC reaches its lowest level.
- Diminishing marginal product is a short-run idea; economies and diseconomies of scale are long-run ideas.

# The essential cost relationships

- Economic profit equals accounting profit minus implicit costs.
- Fixed cost stays fixed; average fixed cost falls.
- Diminishing marginal product causes marginal cost to rise in the usual short-run model: MC = wage ÷ MP.
- Marginal cost below an average pulls that average down, and marginal cost above it pushes that average up. Hence, on the standard U-shaped curves, MC crosses AVC and ATC at their separate minima, from below.
- In the long run the firm can change every input and chooses the lowest feasible average cost at each output: LRAC. Falling LRAC is economies of scale; rising LRAC is diseconomies of scale.